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Assessing ecological changes in and around marine reserves using community perceptions and biological surveys

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ABSTRACT

1. Well-enforced partial or total no-fishing zones (collectively known as marine protected areas, or MPAs) can help restore degraded coral reefs and enhance fish populations.
2. A comparison was made of community perceptions of ecological changes in an MPA with concurrent scientific data on these changes in the same MPA. Such analyses are particularly important in community-based MPAs where local support is a key determinant of ecological success.
3. The no-take MPA in question was initially launched in partnership with the community in 1995 and formalized in 1998. The perceptions data come from interviews with community members in 1999 and 2004, the biological data come from underwater visual censuses of the MPA from 1998 to 2004.
4. Community members perceived more fish within the MPA and slight increases in catch outside the MPA. In contrast, fish censuses showed a high degree of stochastic variation and only minor increases in fish abundance, size and diversity in and around the MPA between 1998 and 2004.
5. Possible explanations for these discrepancies include different temporal, spatial or species frames of reference and/or limitations to the biological survey technique. Other options include wishful thinking, external influences, a desire to please, or confounding with other benefits.
6. This study demonstrates some of the strengths and weaknesses of community perceptions and biological data. In order to improve our understanding about the changes that occur over time in an MPA and engender community support for the long-term viability of MPAs, it is important to develop diverse and efficient monitoring schemes. Copyright © 2010 John Wiley & Sons, Ltd.

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INTRODUCTION

Successful marine protected areas (MPAs), including no-take reserves, can help restore degraded coral reefs and improve the livelihoods of coastal communities by increasing fish size and abundance in adjacent areas (Halpern and Warner, 2002). In the past three decades numerous MPAs have been created as a favoured conservation strategy. However, in many countries, insufficient government funds for enforcement and management of MPAs and poor support from local communities have limited the effectiveness of many reserves (Alcala, 1998; Williams, 1998; White *et al.*, 2002; Mora *et al.*, 2006; Samoilys *et al.*, 2007). Community-based initiatives are seen as a key approach to enhance both the number and the long-term viability of MPAs (Johannes, 2002). These types of reserves involve local communities to establish, enforce and manage the MPA and therefore tend to require less government funds for enforcement and attain higher local compliance (O'Riordan and Stoll-Kleemann, 2002; Mascia, 2003; Lundquist and Granek, 2005). In addition to such practical benefits, there are also ethical arguments towards conservation interventions that have greater support from the local communities (Brandon and Wells, 1992; Wilshusen *et al.*, 2002).

The long-term sustainability of community-based MPAs depends largely on local people believing that the reserve is effective in enhancing fish populations (Suman *et al.*, 1999; Walmsley and White, 2003). Scientists tend to consider biological monitoring data as the only evidence required to

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demonstrate to communities that there has been a change in fish abundance or catch per unit effort over time in and around an MPA (Danielsen *et al.*, 2005; Stem *et al.*, 2005). Such data may, however, be very hard for most communities to obtain, because they lack the funds or technical capacity (Kulbicki *et al.*, 2007; McClanahan *et al.*, 2007b). Even where external organizations do the sampling, there may be difficulties in providing the data rapidly or in the form that the community needs. Some communities undertake a simplified version of structured or participatory monitoring for themselves, but many find it difficult to sustain such initiatives (Danielsen *et al.*, 2005; Rijsoort and Jinfeng, 2005). In reality, many communities — or members of the community — may develop their own perceptions of change, based on a variety of experiences and discussions (Huntington, 2000; Oba and Kaitira, 2006).

Our best chance of understanding the changes in an MPA — and in engendering the community support to sustain it — lies in drawing on and reconciling information from diverse streams of evidence (Mangel *et al.*, 1996; Ferguson and Messier, 1997). It would be a mistake to assume that biological monitoring is any more objective and relevant than community perceptions. Scientific research can be itself value laden and influenced by researcher perceptions in dimensions ranging from sampling regime to input variables for models. Although biological monitoring allows researchers to document exactly how they evaluated changes in fish counts, it is not necessarily true that a more objective fish count will yield a more accurate impression than a subjective look-see by an experienced observer (Huntington, 2000; Mackinson, 2001). On the other hand, surveyed perception responses may be influenced by a suite of social, psychological or economic factors and it is often difficult to determine how communities formed their perceptions of ecological change (Festinger, 1957; Ferguson and Messier, 1997; Ferraro *et al.*, 1997; Innes *et al.*, 1998; Saenz-Arroyo *et al.*, 2005). It is, however, clear that diverse approaches can complement each other in ways that may help advance our understanding of how MPAs operate, and why communities might support them.

This study examined a community-based reserve in the Philippines. In the 1980s, a change in Philippines fisheries policy gave municipal government greater jurisdiction over marine resources to 15 km offshore. By establishing a municipal ordinance, communities can legally enforce fishing regulations or prohibit fishing in their own MPAs (Russ *et al.*, 2004). This change in policy, as well as support from governmental and non-governmental organizations, led to the creation of over 300 community-based marine reserves (Christie *et al.*, 2002).

This paper examines the ecological changes that occur in and around a community-based no-take marine sanctuary in the Danajon Bank, a region with particularly high levels of diversity and intensive human threats to marine ecosystems. In Danajon Bank, villagers began to create their own MPAs because of the high incidence of illegal and destructive fishing practices such as dynamite or cyanide fishing, awareness of local fishers of their own declining fish catch, and a lack of enforcement of fishing regulations by government personnel (Alcala, 1998). Villagers hoped that these MPAs would restore coral reef communities degraded by destructive fishing so that they and future generations could profit from the export of larvae or the spillover of fish from the MPAs (Russ and Alcala, 1996; Côté *et al.*, 2001; Roberts *et al.*, 2001; Russ *et al.*, 2004). To that end, they bore the costs of establishing and maintaining the MPA (Alcala, 1998; Walmsley and White, 2003), albeit with non-governmental support.

The goal of this study is to assess how well biological survey data coincided with local community perceptions of ecological changes in the Handumon sanctuary. We hope that these two types of data will provide complementary information to create a more complete picture of the ecological changes that occur over time in a community-based reserve. Moreover, we will identify the strengths and limitations of two forms of data collection in order to inform strategies for effective and sustainable MPAs.

METHODS

Study Area

The Danajon Bank is a 145 km long double-reef system of shoals and atolls located primarily along the coast of Bohol province. The study was conducted in Handumon village (10°10.3 N, 124° 10.6 E) in Jandayan island, which is a terrestrial island fringed with coral lying to the east of the inner reef of Danajon Bank (Figure 1). The majority of the approximately 1100 villagers are fishers and seaweed farmers. Other sources of livelihood include agriculture, maintaining small stores, selling and transporting freshwater, handicraft, and grouper aquaculture (Panes, 2006b).

History of the MPA

In 1994, as part of a community-based marine conservation programme on seahorses, Filipino biologists and social workers (in consultation with ACJV) began promoting the establishment of a no-take MPA within the community waters. After deliberation and discussion, the villagers allocated one-third of the waters immediately around their village to an enforced, no-take sanctuary. The 50 ha no-take marine sanctuary extended out from the mangroves to protect a linear reef running parallel to the shoreline, and the soft-bottom habitat and deeper waters beyond it. The area had reportedly been rich in marine life but by 1995 it had been severely depleted and damaged by overexploitation and destructive fishing, to the point where there were very few fish of any size or species (personal observations, ACJV). The village council formally recognized the MPA and began enforcing its boundaries in 1995, but the municipal ordinance (Getafe, Bohol) was only signed in August 1998. The Filipino professionals were the precursor of a group that became the Project Seahorse Foundation for Marine Conservation (PSF), a Philippines non-governmental organization that ACJV helped found in 2003. It has close links with technical advisers in international universities and engages in formal research through such collaborations.

Current status

Based on a standardized regional assessment of effectiveness of MPAs in 2004 that incorporate factors such as enforcement capacity, financial management and educational outreach, this MPA had the highest management rating of 19 MPAs in the region (White *et al.*, 2006; Samoilys *et al.*, 2007). Handumon

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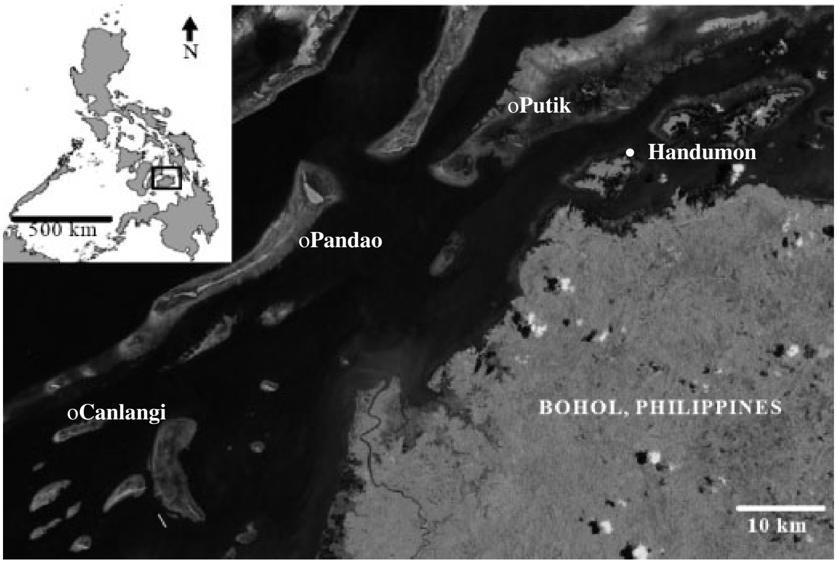

Figure 1. Map of the study area.

has an active and cohesive community organization and management committee as well as frequent engagement with Filipino biologists and social workers from PSF, reinforced by visits from international colleagues. These factors have led to MPA education campaigns, strict enforcement of the MPA, access to external funds for management and high community support for the MPA (Panes, 2006a; Samoilys *et al.*, 2007). In 2007, the Handumon MPA was recognized by the MPA Support Network, a national marine protected areas organization, as the best MPA in the Philippines (Gamolo, 2007).

Community perceptions surveys

The questionnaires were conducted in the local language (Cebuano) from (a) 20 March to 30 April 1999 and (b) 13 to 18 March 2004. The two Filipino professionals who conducted the surveys (Rosemarie Apurado, a community organizer, and Marivic Pajaro, a biologist) worked for PSF and each had years of experience of community work in Handumon. Each interview lasted approximately 30 min. Fifty-nine villagers were randomly selected to participate in the survey in 1999 and 37 of the original participants were resurveyed in 2004; the others had moved away from the region, died or were not available on the dates of the surveys.

Community members varied in the personal experience they could call on to assess temporal changes in fish and fisheries catch, depending on their daily activities.

Respondents were first asked if they currently swam in the reserve or were fishers. Swimmers provided an assessment of changes in fish abundance, size and diversity within the reserve area. Swimmers also indicated whether they had swum in the area of the reserve before the MPA was established. Swimmers who had swum in the reserve area prior to establishment and thus were likely to have greater personal longitudinal experience are called ‘continuing swimmers’ in this study. This was a coarse grouping because continuing swimmers may not always have more personal experience than other community members who did not have experience swimming in the MPA before the establishment of the MPA. The level of personal experience would depend on factors such as how often and how long swimmers swam and whether they used a swimming mask, as well as whether non-swimmers (during the survey period) had previous long-term and extensive swimming experience in the reserve.

Fishers provided assessments of changes in total number of fish caught, size of fish caught, number of fish species caught outside of the sanctuary and income generated from fishing. Fishers were also asked if they currently fished in the same location as they did before MPA establishment. Fishers who had fished in the same location before MPA establishment are called ‘continuing fishers’ in this study.

Respondents who were both swimmers and fishers commented on both fish conditions inside the MPA as well as fish catch outside the MPA. All respondents were asked to rate whether these fish or fisheries catch variables had changed, in one of the following five categories: up a lot, up a little, unchanged, down a little, or down a lot in 1999 and 2004 compared with conditions immediately before the MPA was established (in 1995).

People who were neither swimmers nor fishers (‘non-swimmer/non-fisher’) during the time of the survey also provided their perceptions on changes in fish abundance, size and diversity inside the reserve area (Table 1, for sample sizes). For these respondents, it was not determined how much personal experience they may have had to assess temporal changes within the reserve. For example, a respondent who swam from 1980 to 1998 in the reserve area would be considered a non-swimmer for the survey, but would have had longitudinal personal experience. In contrast a non-swimmer could also be a terrestrial farmer who would have never swum in the MPA and would have no longitudinal personal experience.

Visual fish census

The underwater visual census (UVC) was conducted using standardized methods that are described in detail elsewhere (Samoilys and Carlos, 2000; Samoilys *et al.*, 2007). Four, 50 by 5 m permanent belt transects were established parallel to the shore along the reef, both in Handumon MPA and

Table 1. Sample sizes of community members who responded to surveys on fish or fisheries related changes inside and outside the MPA for 1999 and 2004.

Survey Responses:	Inside MPA (fish)				Outside MPA (catch)				Inside and outside
Current Activities:	Swimmers		Non-swimmer		Fishers		Non-fisher		Fisher + Swimmer
Time period:	Continuing	No pre-establishment	Pre-establishment	Never	Continuing	No pre-establishment	Pre-establishment	Never	
1999	19	8	27	4	21	4	1	5	31
2004	7	0	17	5	15	2	2	0	18

'Current Activities' refers to the activities of the respondent during the year of the survey. 'Fisher + Swimmer' refers to fishermen who were also swimmers and had commented on changes inside the MPA. 'Continuing' swimmers and fishers have pre-MPA establishment personal experience. 'Pre-establishment' refers to respondents who were not swimming or fishing at the time of the survey, but swam and fished before MPA establishment.

immediately outside the MPA. Four more transects were surveyed in three sites distant from Handumon, and thus beyond the direct influence of the MPA, at 4 km (Putik), 15 km (Pandao), 30 km (Canlangi) from Handumon (Figure 1). Fishers from Handumon reported that they frequently fished at Putik, occasionally went as far as Pandao, and certainly had interaction with fishers from other communities who would have reported catches even further than 15 km away.

UVCs were conducted twice per year, once in the dry season (Feb–May) and once in the wet season (Aug–Nov) between 1998 and 2004 by a team of trained Filipino biologists and trained international volunteers. During the census, one biologist and one volunteer swam along the transect, identified the fish to family level (for expediency), counted all individuals and recorded the approximate sizes to the nearest centimetre. Each transect took 15 min and at the end of the transect the biologist and volunteer compared data sheets to ensure that they did not recount any aggregations of fish crossing the transect.

Statistical analyses

Community perceptions

Rating responses were converted to ordinal scores as follows: up a lot = 0, up a little = 1, no change = 2, down a little = 3 and down a lot = 4. Due to the high number of people who responded 'up a lot' (ordinal score = 0), the scores followed a Poisson distribution (as tested by a Kolmogorov–Smirnov test) and so a generalized linear mixed effects model (GLMMpql, package MASS), specifying Poisson distributed error terms, was used to simultaneously test for multiple effects while controlling for the random effect of respondent (37 of the same respondents were surveyed in both years). The responses of the swimmers and fishers who had no pre-establishment experience were combined with non-swimmer/non-fishers as it was not possible to assess the longitudinal personal experience level for these groups based on the surveys and they are likely to have had less personal experience than the continuing swimmers or fishers. For the models predicting fish characteristics within the reserve the effect of year was included, whether or not they were continuing swimmers and whether or not they were fishers. Continuing swimmers were included as an effect in the model in order to assess whether respondents with possibly greater personal experiences had differing perceptions of ecological changes. Fisher or non-fisher was also included in the analysis because fishers may have different perceptions of change within the MPA as they are more likely to compare the state of the ecosystem within the MPA with general conditions outside of the MPA.

For the model predicting catch characteristics outside the reserve the effect of year and whether or not respondents were 'continuing fishers' were included.

Underwater fish census

Data on fish abundance, fish size and fish family diversity for underwater fish censuses were normally distributed after log transformations.

Fish abundance. For fish abundance all fish families were examined first and then the analysis was restricted to only top and mid trophic level and large herbivore fish families: Acanthuridae (herbivore), Balistidae (mid trophic), Haemulidae

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(mid trophic), Holocentridae (mid trophic), Lethrinidae (mid trophic), Lutjanidae (mid-top trophic), Mullidae (mid trophic), Nemipteridae (mid trophic), Scariidae (herbivore) and Serranidae (mid-top trophic). Although almost all fish families are caught by fishers in Handumon (Samoilys *et al*., 2007), these fish families respond more to protection and have life history traits that make them more vulnerable to population declines owing to fishing (Jennings *et al*., 1999; Côté *et al*., 2001; Micheli *et al*., 2004; Barrett *et al*., 2007). These fish also have particularly high economic importance (Samoilys *et al*., 2007) and so are more likely to influence villagers' perceptions than less valuable fish families.

To assess whether fish abundance had changed over time linear mixed effects models (lme, package lmne, R) on log-transformed data [log10 (x– 1)] were used. In the model, Year and Season (Wet and Dry) and their interactions as fixed effects were included, as well as transect identity as a random effect. Season was included in the model because variability between wet and dry seasons during the same year could mask possible temporal trends. Intra-annual variability may be particularly high for longer lived, larger fish that have episodic recruitment and emigration out of the reserve. This variability could have masked inter-annual changes in fish abundances (Barrett *et al*., 2007). Transect number was also included in the model because differences in the recovery response could also possibly be attributed to minor variations in location of the transects (Benedetti-Cecchi *et al*., 2003).

Restricted maximum likelihoods (REML) was initially used to calculate log ratios because they are less biased in their calculation of variance components than ML (Pinheiro and Bates, 2000), but for model comparisons maximum likelihoods (ML) were used. Insignificant terms were dropped and significant differences of the AIC value were tested for in different models in lme (package lmne, R).

As the objective of this study was to compare biological results with social interview results, the data for the four transects within the MPA were analysed separately from the fish data for the four transects outside the MPA. A previous study included both data from inside and outside MPAs in order to evaluate the effectiveness of MPAs. This previous study examined the interaction terms between year and treatment (inside and outside reserve and control) (Samoilys *et al*., 2007).

Fish size. The analysis on temporal changes in fish size was restricted to mid and top large trophic fish which were frequently observed within the reserve: Labridae, Lutjanidae, Nemipteridae, Scariidae, Serranidae. Although many other large fish families were observed in Handumon, at least one individual of each of these five families was observed in more than half of the sampled transects, allowing for greater analytical power.

During a census, fish sizes for a particular family may have been recorded multiple times. To minimize pseudoreplication, the data were first aggregated to give only one weighted mean fish size per family for a census before using the same linear mixed effects modelling approach as above. The data set was separated by family and separate analyses were run testing only for the effects of year and season and the interaction between these two variables.

Family diversity. Fish family diversity was examined on the basis of family richness (total number of families) (Colwell *et al*., 2004). The linear mixed effects model was conducted using the same methods described for fish abundance.

RESULTS

Community perceptions survey

The majority of respondents ($n_{1999} = 50$, $n_{2004} = 29$) said that the fish abundance, size and diversity inside the reserve was 'up a lot' and mixed effects models suggested that neither year (1999 and 2004), nor whether or not respondents were continuing swimmers or fishers had a significant effect on the responses to changes in most fish and catch characteristics (Table 2). The responses of the same individual for fish abundance, size and diversity within the reserve in the same year were correlated with each other (Pearson's $r_{1999} = 0.69$, Pearson's $r_{2004} = 0.75$). However, it appeared that continuing swimmers had a slightly more positive perspective of increases in fish size than people who did not have long experience (Figure 2), whereas there were no statistically significant effects for abundance and diversity within the reserve. In addition, fishers appeared to be less likely than non-fishers to state 'up a lot' for changes in fish size inside the MPA (Figure 2).

Table 2. Results of generalized linear mixed effects models testing how year (1999 and 2004), personal experience level (continuing swimmer/fisher or other) or whether or not respondent was a fisher during the time of the survey affected community perceptions of changes inside and outside the reserve.

	Year			Personal experience level		Fishers/non-fishers	
	T	Df	*P*	*T*	*P*	*T*	*P*
Inside reserve							
Abundance	−0.44	24	0.662	1.2	0.238	−2.0	0.051
Size	0.309	25	0.76	−2.9	**0.008**	2.48	**0.020**
Diversity	0.014	23	0.796	−2.0	0.06	−0.063	0.950
Catch outside reserve							
Abundance	–1.7	11	0.11	–0.08	0.94		
Size	–1.9	11	0.08	0.15	0.88		
Diversity	–1.7	7	0.13	0.34	0.73		
Income	1.2	10	0.24	–0.72	0.48		

For personal experience level inside the reserve, respondents were either continuing swimmers, or, others, who could be swimmers with no pre-MPA swimming experience or non-swimmers. For catch data outside of the MPA, respondents were categorized as continuing fishers or people who did not have personal longitudinal pre-MPA fishing experience according to the surveys.

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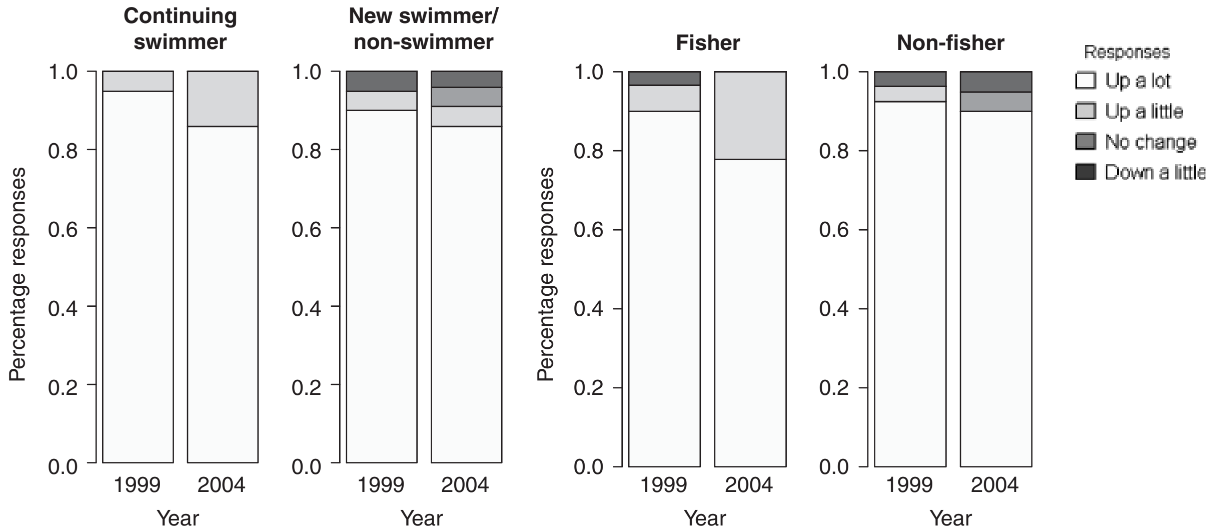

Figure 2. Community perceptions on changes in fish size since the establishment of an MPA. 'Continuing swimmer' swam during the time of the survey as well as pre-MPA establishment while 'New swimmers' began swimming some time after MPA establishment. No respondents stated 'Down a lot' for any of the survey questions. Responses regarding fish abundance and diversity showed similar patterns to size.

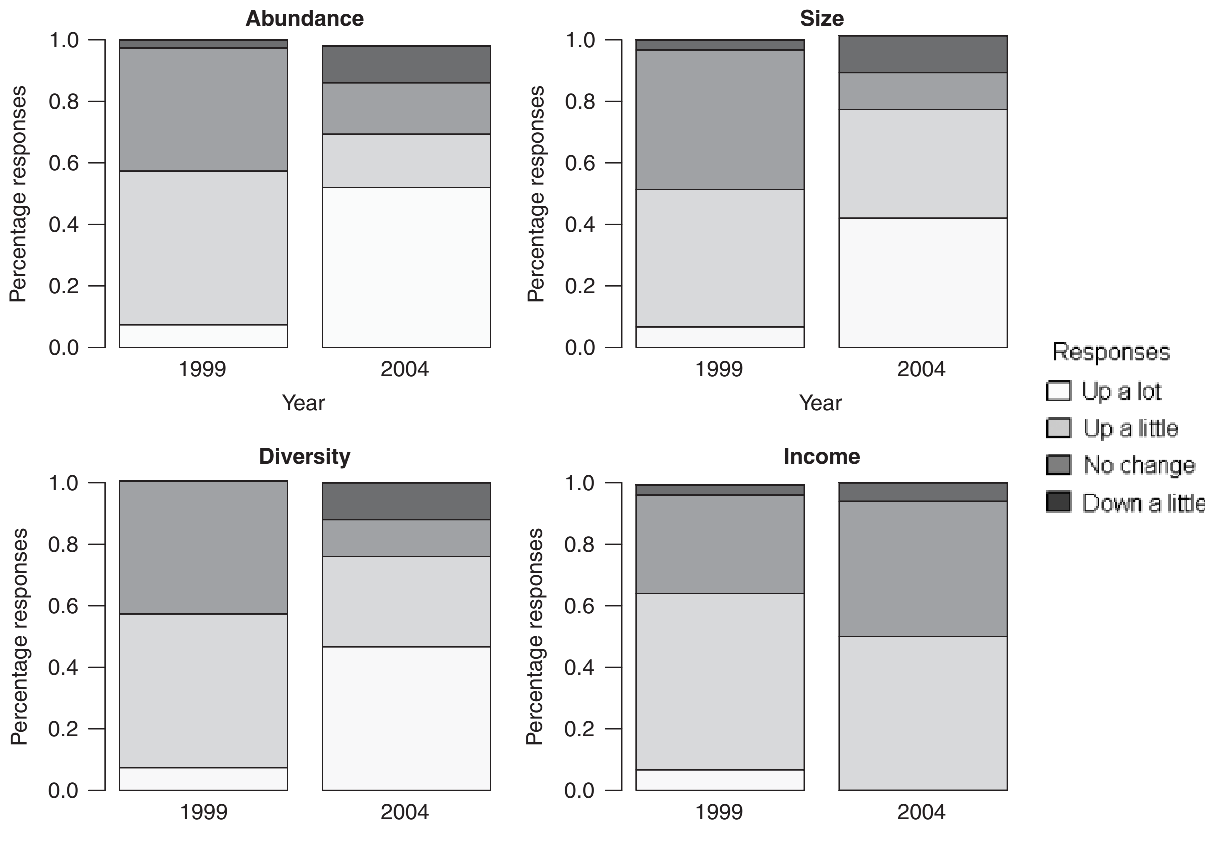

Figure 3. Fisher perceptions of changes in catch abundance, size, diversity and income from fishing outside of the MPA compared with before establishment of the MPA. For this graph, fishers who may or may not have longitudinal personal experience were pooled because mixed effects models suggested that there were no significant effects on responses. No respondents stated 'Down a lot' for any of the survey questions.

In general, Handumon villagers were more positive about changes in total fish catch, catch size, catch family diversity and catch income inside the MPA compared with changes in fish abundance, size and diversity outside the MPA (Figure 2 cf. Figure 3).

Most respondents who were interviewed in both years and who commented on changes in fish stocks within the sanctuary gave the same response in 1999 and 2004. Most fishers who were surveyed in both years had a more positive perspective of fish catch in 2004 than in 1999, although the difference was not significant (Table 3).

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Table 3. Frequencies of changes in fish stock perceptions for repeated surveyed respondents between 1999 and 2004.

	More positive in 1999	More positive in 2004	No change
Inside reserve			
Abundance	5 (17%)	3 (10%)	21 (72%)
Size	4 (14%)	3 (10%)	21 (75%)
Diversity	4 (15%)	3 (12%)	19 (73%)
Catch outside			
Abundance	4 (28%)	8 (57%)	2 (14%)
Size	3 (21%)	8 (57%)	3 (21%)
Diversity	2 (20%)	6 (60%)	2 (20%)
Income	5 (39%)	3 (23%)	5 (39%)

'More positive in 1999' indicates that the respondent had a lower ordinal response score in 1999 than in 2004. Ordinal score were: (0 = Up a lot, 1 = Up a little, 2 = No change, 3 = Down a little, 4 = Down a lot). 'No change' indicates that the respondents gave the same perceptions level in 1999 and in 2004.

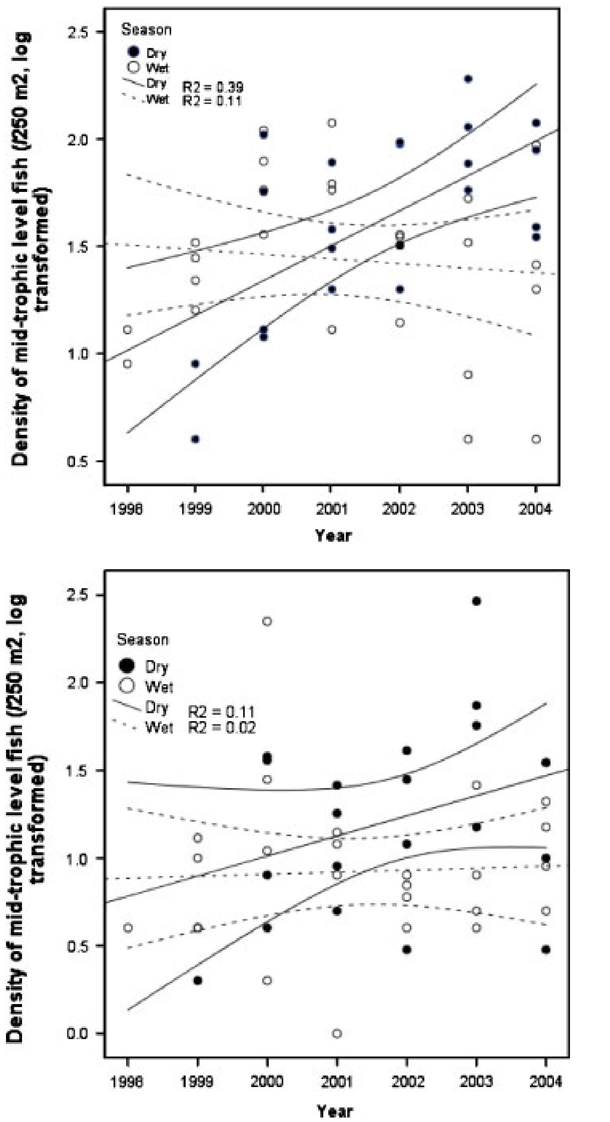

Figure 4. Density of mid-trophic level fish families inside (top) and immediately outside (bottom) Handumon reserve over time. Year had a statistically significant effect only for transects inside the reserve. Lines represent 95% confidence intervals.

Underwater visual census

Results demonstrated that year had a significant positive effect on total fish abundance, mid-trophic level fish abundance and the total number of families observed inside the reserve, but not immediately outside the reserve (Figure 4, Table 4). Labrids (54%) and scariids (36%) were the main drivers for the positive significant effect in mid-trophic level fish numbers inside the reserve. Year had no significant effect on top-trophic fish abundance inside, outside and at reference sites. There was a significant increase in the total number of families in the reference sites over time, but year did not have a significant effect for other parameters in the reference sites.

There was no significant temporal change in fish sizes for the most frequently observed common large fish families within the reserve, and immediately outside of the reserve (Table 5, Figure 5). However, there appeared to be a decline in the size of labrids, scariids and serranids in the reference sites (Figure 5). The effect of year on fish size varied across sites for labrids and between seasons for scariids.

DISCUSSION

Results indicated a gap between community perceptions and biological surveys. While villagers perceived significant improvements in fish stocks, biological data did not demonstrate substantial or consistent temporal changes in fish abundance, size and diversity within or immediately outside of the MPA. Similar results have been demonstrated in previous studies (Côté *et al.*, 2001; Walmsley and White, 2003). A previous study also suggested that fishing communities can strongly support MPAs and believe in their effectiveness at improving ecological conditions, even if there are no major ecological changes detected by biological studies (Walmsley and White, 2003).

Most studies on community-based MPAs have focused on using biological data (whether collected in a participatory manner or by outside professionals) to convince local people of the MPA's effectiveness (Stem *et al.*, 2005). Here the challenge is of a different order: how do we explain the relative optimism of stakeholder perception?

Community members may have had more positive perceptions than the biological results because of (a) a longer temporal reference, (b) a broader spatial reference, (c) a more specific taxon reference, (d) more thorough sampling, (e) wishful thinking, (f) external influence, (g) a desire to please, or (h) confusion with other benefits.

First, the collection of systematic biological data began in 1998 (three years after MPA enforcement began) whereas community respondents were comparing 1999 and 2004 conditions to pre-MPA establishment (1995). In some study areas, population recovery of smaller fish may be most evident within the first 3 to 8 years after the establishment of the reserve (Roberts *et al.*, 2001; Walmsley and White, 2003; Barrett *et al.*, 2007), although most studies show slower recovery rates and variability across families (Russ and Alcala, 1996; McClanahan *et al.*, 2007a).

Second, community members may have used a different reference point, that of the surrounding waters, rather than the initial state of Handumon's MPA. Previous reviews in this region have documented large scale regional decline within the

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Table 4. Results of linear mixed effects models on factors affecting fish abundance and diversity in and immediately around reserves and in distance reference sites.

			B	*DF*	*T*	*P*
Total fish abundance						
	Inside	Year	0.07	42	3.0	**0.0042**
		Season	−0.20	42	−2.7	**0.0103**
	Outside	Year	0.07	41	1.9	0.071
	Control	Site (Canlangi)	0.55	9	10.1	<**0.0001**
		Site (Pandao)	−0.03	9	−0.5	0.63
Top-trophic abundance						
	Inside	Season	0.12	43	1.1	0.30
	Outside	Season	−0.19	41	−1.9	0.062
	Control	Year	−0.03	145	−1.8	0.07
		Site (Canlangi)	0.28	9	3.6	**0.006**
		Site (Pandao)	−0.33	9	−4.1	**0.002**
Mid-trophic abundance						
	Inside	Year	0.33	41	3.34	**0.0018**
		Season	362	41	3.1	**0.0036**
		Year × Season	−0.18	41	−3.1	**0.0036**
	Outside	Season	−0.30	41	−2.4	**0.019**
	Control	Year	0.025	145	1.9	0.06
		Site (Canlangi)	0.33	9	3.6	**0.006**
		Site (Pandao)	−0.16	9	−1.7	0.138
Family richness						
	Inside	Year	0.604	43	2.56	**0.014**
	Outside	Year	0.351	40	1.7	0.096
		Season	−1.6	40	−2.4	**0.024**
	Control	Year	0.26	145	2.24	**0.027**
		Site (Canlangi)	−0.78	9	−1.4	0.204
		Site (Pandao)	−3.6	9	−6.1	**0.0002**

Shown are the results of the final models based on AIC likelihoods. Total fish abundance, top-trophic and mid-trophic level fish abundance was log transformed. Test values represent differences in sites relative to Putik (the reference site closest to Handumon) and for season, relative to the dry season.

Philippine fisheries (Christie *et al.*, 2007). The three reference sites distant from the MPA demonstrated a decline in fish size for three large food fish families (Labridae, Scariidae and Serranidae). There is little doubt that severe fishing pressure and other external factors such as climate change continue to deplete fish stocks outside MPAs (Boersma and Parrish, 1999; Halpern, 2003), perhaps making an MPA appear to do rather better in relative terms than it is in absolute terms.

Third, villagers might have classified and evaluated fish differently from the surveys. Family level observations and analyses were used, whereas villagers may well have drawn their conclusions from particular species or size classes that have considerable economic, dietary or cultural importance. Moreover, it would be entirely possible for numbers in any given family to remain constant, at least statistically, while one or more species were responding particularly well to the MPA. Studies in other regions have demonstrated significant differences in recovery rates to protection among families or species with different life histories (McClanahan *et al.*, 1999, 2007a; Côté *et al.*, 2001).

Fourth, high natural stochastic variability in fish counts and fish surveys may have masked trends in the biological surveys. Fish frequently move in and out of transects and individuals that were not documented in the census could still be within the MPA (Kulbicki and Sarramegna, 1999; Kulbicki *et al.*, 2007; McClanahan *et al.*, 2007b). The biological surveys followed international protocols for coral reef monitoring, maintained standardized protocols for seven years, and were analysed with mixed effects models to account for some of the intra-year variability. Nonetheless, the biological surveys occurred only twice a year, for a total of 1 h (four transects × 15 min) of sampling each time. Greater sample sizes per sampling period and more frequent sampling would have enhanced the statistical power and increased the likelihood of detecting statistically significant differences. Swimmers and fishers who 'sample' the fish densities more frequently and without the constraint of a 5 m transect may be better able to integrate short-term stochastic variation and detect longer-term trends.

Fifth, villagers may be optimistic because of their commitment to the MPA process, in what might be termed wishful thinking. Cognitive dissonance theory suggests that voluntary changes in behaviour can lead to changes in beliefs that serve to justify those behavioural changes (Festinger, 1957; Gringart *et al.*, 2008). Villagers may have adjusted their beliefs to perceiving that the MPA was improving fish stocks because they had voluntarily invested time and effort in selecting, establishing, and managing the MPA.

Sixth, villagers may be responding to outreach campaigns by PSF, the group that catalysed the establishment of the

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Table 5. Results of linear mixed effects models on factors affecting log-transformed fish length (cm) from visual fish census surveys.

			B	*DF*	*T*	*P*
Labridae						
	Inside	Season	−0.027	50	−0.76	0.45
	Outside	Year	0.0108	30	0.97	0.34
	Control	Year	−0.017	123	−1.6	0.1107
		Season	−0.087	123	−2.5	**0.0126**
		Site (Canlangi)	60.5	9	2.1	**0.0638**
		Site (Pandao)	−24.7	9	−0.8	0.4291
		Season × Site(Canlangi)	0.043	123	0.9	0.3619
		Season × Site (Pandao)	0.15	123	3.0	**0.0031**
		Year × Site(Canlangi)	−0.03	123	−2.1	**0.0364**
		Year × Site(Pandao)	0.01	123	0.82	0.411
Lutjanid						
	Inside	Season	−0.14	15	−1.5	0.16
	Outside	Season	−0.07	20	−1.2	0.25
	Control	Year	0.047	11	1.8	0.11
Nemipteridae						
	Inside	Year	0.011	34	0.75	0.45
	Outside	Season	0.11	38	2.4	**0.019**
	Control	Year	0.0187	45	1.1	0.28
Scariidae						
	Inside	Season	−0.11	20	−1.7	0.097
	Outside	Year	−0.02	30	−0.8	0.40
	Control	Year	−0.05	104	−3.5	**<0.0001**
		Season	−118	104	−3.5	**<0.0001**
		Year × Season	0.05	104	3.5	**<0.0001**
Serranidae						
	Inside	Season	−0.06	27	−1.1	0.29
	Outside	Season	−0.10	21	−1.4	0.18
	Control	Year	−0.04	81	−3.6	**0.0005**
		Season	−0.11	81	−3.0	**0.0032**

Shown are the results of the final models based on AIC likelihoods. Test values represent differences in sites relative to Putik (the reference site closest to Handumon) and for season, relative to the dry season.

MPA, and has remained active in this and many other resource management and capacity building endeavours in Handumon. Initially PSF facilitated a site visit for Handumon villagers to the highly successful reserve at Apo Island (Russ and Alcala, 1996; Walmsley and White, 2003). Thereafter, PSF has organized environmental educational campaigns, capacity building workshops and meetings with community members (Lavides *et al.*, 2005). Such ventures could have led to a 'priming effect' whereby prior experiences subconsciously affect people's response to a subsequent event (Simon, 1988; Ober *et al.*, 1991). It is argued that a villager who has been told that an MPA will increase fish abundance may be more likely to detect and remember a swim in which they saw a high number of fish in the water (Simon, 1988; Ober *et al.*, 1991).

Seventh, villagers could have been reporting what they thought the researchers wanted to hear. The presence of PSF (and its staff house) in Handumon offers some, albeit limited, economic opportunities and helps raise the profile and prestige of the community, to the extent that villagers might have offered strategic answers in order to keep the PSF team engaged locally. We like to think that the past experience of the PSF community organizers in Handumon could have helped elicit honest responses but villagers might also have offered information influenced by motivational biases and strategic behaviour. That said, the consistency of answers across all respondents makes this explanation less likely, not least because admission of problems with the MPA might arguably be a better strategy for enhanced NGO involvement.

Finally, villagers may have been positive about ecological changes because they are experiencing a psychological lift from undertaking any positive resource management activity. The ongoing community activities related to the management of the MPA may have helped instill confidence, hope and a feeling of greater control over the future (Rotter, 1966). For a poor and disempowered community that initially had almost no fish within the reserve area, a 'small' change as detected by biologists may have a significant social or psychological importance. It is becoming clear that MPAs can offer benefits in terms of social and political capital far in excess of the documented biological changes (Duram, 1997; Pretty and Smith, 2003).

Given that MPAs will certainly continue to be a strong tool in marine conservation (Agardy *et al.*, 2003), it is vital that the best assessment is made of their effectiveness, using all available analytical approaches. Neither community perception nor biological surveys will provide a comprehensive picture of MPA effectiveness. Rather, all available information needs to be collected, such that findings can be cross-checked and challenged (Huntington, 2000; Mackinson, 2001). Related analyses on the same data set — but focused on comparing inside, outside and distant sites — actually suggested that there was indeed some biological recovery within the Handumon MPA, with higher fish abundance and significant differences in community composition inside and outside the MPA (Samoilys *et al.*, 2007). On the other hand, the consistent and considerable optimism among respondents needs more careful analysis. Much remains to be learned about psychology and group perceptions in such situations (Miranda *et al.*, 1997; Topp-Jørgensen *et al.*, 2005; Saunders *et al.*, 2006).

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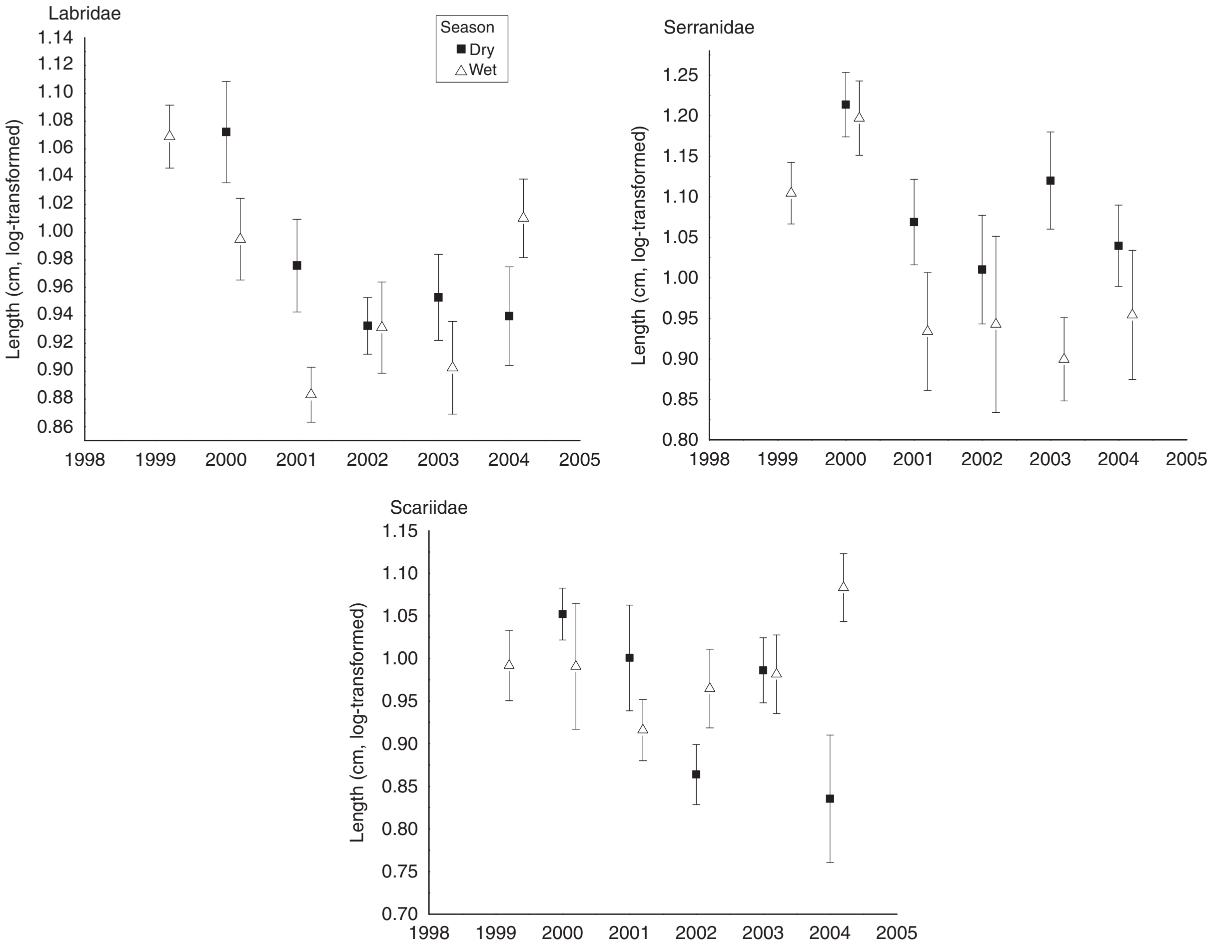

Figure 5. Log-transformed fish lengths obtained from underwater visual censuses for labrids, scariids and serranids over time in reference sites. Shown are standard errors of means pooled across three reference sites (Putnik, Canlangi and Pandao).

It is important to understand the size and scale of benefits needed in order to generate community support (Polansky, 2003). This study suggests that relatively modest biological gains may generate considerable optimism. The challenge is to determine which explanation might best justify the enthusiasm that people of Handumon apparently feel for their MPA, and how transferrable such support might be with other MPAs that have been only recently established or are recovering slowly. Such information would help managers decide which comparisons matter most to local people, how survey techniques might need to change, which messaging is most reliable, how relationships influence MPA support, and what broader benefits are valued. Any optimism based largely on information from the external catalyst needs to be carefully managed to avoid later disappointment, cynicism and poor relationships if recovery does not happen quickly (Boonzaier, 1996).

Participatory monitoring is often touted as an important element of MPA management, enabling local people to make their own deductions about change (Pretty, 1995; Danielsen *et al.*, 2003; Rijsoort and Jinfeng, 2005). In general, however, such monitoring is difficult to implement and of short duration, largely because the scientific rigour deemed necessary by the international scientific community is grossly mismatched to the expertise and resources available in fishing communities (Danielsen *et al.*, 2003; Topp-Jørgensen *et al.*, 2005). There is real need for communities to develop monitoring techniques appropriate to their interests and resources. A first step might be to dissect the manner in which community members currently develop their impressions in the absence of such monitoring. Then the community's optimism in Handumon might be better understood.

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